HO3 New Business (Maria Alvarez) — End-to-End Implementation Plan

Sep 28, 2026 · @Arshpreet Singh Bhatia

1. Summary & scope

Maria's HO3 journey (Blueprint §6.1–§6.7) is delivered in **7 phases** by adding **6 new modules** to `app`, **~11 adapters** to `integ`, **4 new Temporal workflows** in `wfc`, and **7 screens** (from the UX Inventory, plus 1 new Rules Manager screen). Every accept / refer / decline decision runs through one config-driven **Coastal Rules Engine**, so Underwriting and Compliance change thresholds without a code deploy.

The running example. Maria Alvarez buys an HO3 on a $500K frame home in Sarasota County, FL, on **SafePoint Insurance Company, admitted** paper. Five-axis key: `SPIC × HO3 × FL × ADMITTED × ALL_PERILS`. For the trace, we assume she has a **mortgage** (exercises the mortgagee party and the flood mandatory-purchase gate) and **pays in full by card**; mortgagee escrow is a designed variant, not the traced path.

In scope: Customer & Policy → Location → Building/Risk → Coverages → Rating & Pricing Summary → Payment → Issuance → IN_FORCE, plus the post-bind inspection loop (Tier 3 of the Coastal Filter).

Out of scope for this plan: endorsement, renewal, cancellation, assumption/takeout; ACORD/AL3 download (D-07); NFIP flood (D-01); GL posting (we record the fee/premium split, we don't post it); Duck Creek coexistence (see `safepoint-duckcreek-migration`).

As-built reality (MYCEL-ARCHITECTURE §4.4–4.8). Today only the renewal-case slice exists: no quote model, rating engine, billing, documents, product config, outbox, or `integ` adapters. Everything below is **net-new** except where marked *reuse*: `UnderwritingReviewWorkflow`, the `uw_decision_recorded` signal, the BFF route table, and the effective-dated config pattern of `notice_rule_set`.

One scope tension to settle first. The Blueprint header says underwriting is out of scope ("never *should we insure*"), while About-SafePoint's Coastal Filter requires knockouts and referrals. This plan resolves it by separating ownership: **Underwriting owns the rule content (config); the PAS owns the engine that executes it.** Logged as proposed decision **D-10** (§12).

2. Maria's journey, end to end

The journey is one quote that passes seven rules checkpoints, is priced by a separate deterministic rating engine, and becomes a policy only when three workflows hand off cleanly.

Every step of the quote passes a configured rules checkpoint

Each checkpoint returns PASS, REFER, REQUIRE, FORCE_TERMS or KNOCKOUT



Maria's journey · 8 steps, 3 workflows, 7 rules checkpoints

Read down the middle column; each step calls its checkpoint on the right, and the band on the left owns the durable process. Rating sits outside the rules engine on purpose: rules decide *eligibility and terms*, rating decides *price*.

#	Step	Screen (tab)	What fires	Integrations called	Checkpoint	Outcome for Maria
1	Customer & Policy	Quote Wizard tab 1	`POST /v1/submissions` creates Quote `DRAFT`; starts `QuoteWorkflow`	LexisNexis identity prefill (optional)	CP1 Offering	Five-axis offering + product version resolved; Maria = Named Insured, lender = Mortgagee (lien 1)
2	Location	Tab 2	Signal `section_updated(location)` → parallel enrichment activities	Geocode, Verisk PPC/BCEGS, flood determination, coast distance / wind pool	CP2 Location	Territory, PPC, BCEGS, flood zone, coast distance saved as facts with provenance; low–high band shown as `PROJECTED`
3	Building / Risk	Tab 3	Signal `section_updated(building)` → risk enrichment	ZestyAI roof, BuildFax permits, LexisNexis CLUE	CP3 Risk	Agent-entered roof year checked against permit + imagery; a conflict → REFER, Bind locked
4	Coverages	Tab 4	Update `RerateQuote`	Verisk replacement cost (from step 2)	CP4 Coverage terms	Cov A $500K checked against replacement cost; hurricane-deductible floor applied if CP2 forced terms
5	Rating & Pricing	Tab 5	`RerateQuote` → `rating.Rate()` activity	—	Rating engine	Technical → net → admitted taxes/fees → total; status `QUOTED`; quote-validity timer starts
6	Accept & Payment	Tab 6 + Quote & Bind Workbench	Signal `quote_accepted` → `BindWorkflow`	OFAC screen, payment gateway	CP5 Pre-bind gates	Card captured; payment ≠ issuance; activity writes `BOUND`
7	Issuance	Tab 7	`BindWorkflow` issuance activities	Document provider, delivery	CP6 Issuance readiness	Policy number, NB `policy_transaction`, dec page to Maria, agent and lender → `ISSUED` → `IN_FORCE`
8	Post-bind	Policy 360	`PolicyLifecycleWorkflow` → child `InspectionFollowUpWorkflow`	Inspection vendor	CP7 Post-bind	Wind-mit inspection ordered; 30-day timer; result re-runs CP7

3. Architecture & ownership

All new domain logic lands as **modules inside `app`** (one Postgres schema each, no cross-schema joins); every vendor call goes through `integ`; every durable step runs in `wfc`. No new Mycel app is needed.

Component	Owns	Schema	Status today
`app/policy`	Party, PartyRole, mortgagee registry, Policy, PolicyTerm, PolicyTransaction (bi-temporal), policy-number sequences, transactional outbox	`policy`	**Extend** — renewal slice exists; additive migrations only
`app/product`	Five-axis offering, ProductVersion, Form, CoverageDefinition, tax/fee rules, quote-validity days, form-to-template map	`product`	New
`app/rules`	Rule sets, immutable versions, parameters, decision tables, evaluator, evaluation log	`rules`	New
`app/quote`	Submission/Quote, location, building, risk facts, coverages, stipulations, pinned rating result — the Blueprint's "Quote DB"	`quote`	New
`app/rating`	Versioned rate tables, pure `Rate()` function, premium breakdown — the "Rate Engine DB"	`rating`	New
`app/billing`	Billing account, payer (insured / mortgagee), payment plan, invoice + lines, payments, fee/premium split	`billing`	New
`app/document`	Templates, generated documents, per-recipient delivery records	`document`	New
`integ`	~11 vendor adapters, vendor call log, inbound webhooks (payment, inspection)	`integ`	Scaffold → build
`wfc`	`QuoteWorkflow`, `BindWorkflow`, `PolicyLifecycleWorkflow`, `InspectionFollowUpWorkflow`; *reuse* `UnderwritingReviewWorkflow`	—	Extend
`bff`	Routes, 3 aggregation endpoints, idempotency dedup storage	`bff`	Extend (dedup store is a Phase 0 prerequisite)

Component	Owns	Schema	Status today
`kpi`	Fact tables fed from the outbox	`kpi`	Deferred — no ClickHouse wired

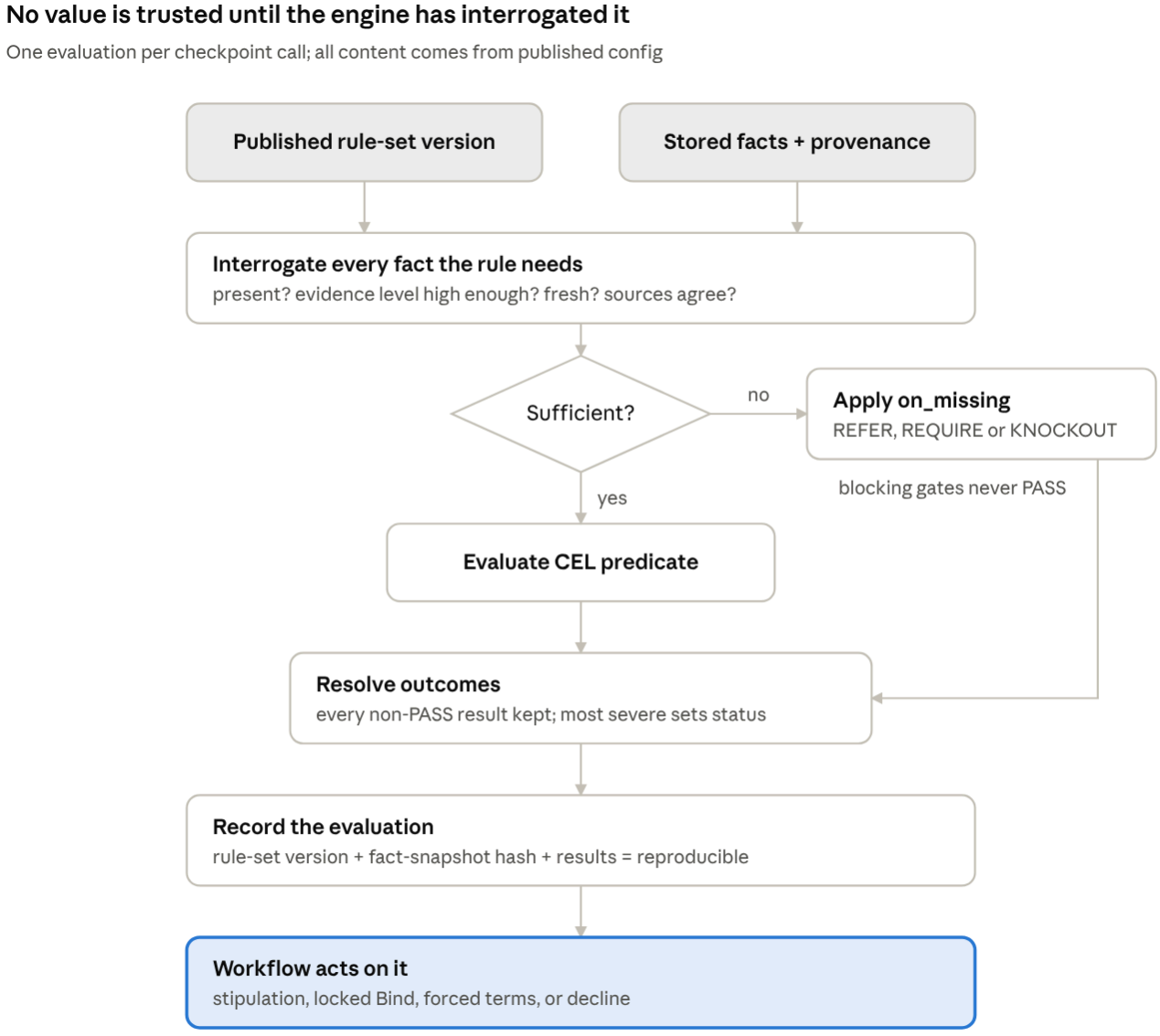
Call path. Svelte `<policy-app>` → BFF (`api.lvh.me` , new screens call the BFF from day one) → `quote` module saves draft data and signals the workflow → `wfc` activities call `integ` , `rules` , `rating` , `policy` , `billing` , `document` → `policy` writes the outbox row in the same transaction.

Design rules that shape everything below:

- **Single writer, widened.** Only a `wfc` activity writes `policy.status` , `policy_transaction` , *and* `quote.status` . Handlers may save draft section data; they never move a status.
- **Rules and rating are both pure, versioned, and separate.** Rules answer *may we and on what terms*; rating answers *how much*. Neither calls a vendor — they read facts already fetched and stored.
- **Facts, not form fields.** Every enriched value is stored as a `risk_fact` with source, confidence and `fetched_at` , so the rules engine can interrogate *where a value came from*, not only what it is.
- **Five-axis key everywhere.** Rules, coverage definitions, rate tables, tax/fee rows and document templates are all scoped by the same `(carrier_paper, product, state, market_type, peril_scope)` key plus effective dates. No `if state == "FL"` anywhere.

4. The Coastal Rules Engine

The engine never accepts a submitted value at face value: before any rule runs, it checks that every fact the rule needs is **present, backed by strong enough evidence, fresh, and consistent across sources**. All rule content — predicates, thresholds, outcomes, messages — is versioned config that Underwriting publishes with an effective date, so changing "roof age 15 → 12" is a publish, not a deploy. Target from About-SafePoint: **60–70% of submissions auto-processed** (instant quote or instant decline).



Rules engine evaluation · interrogate, evaluate, resolve, record

The right-hand branch is what makes it an *interrogation*: missing, stale or conflicting evidence never falls through to PASS; the rule's own `on_missing` decides whether that means a referral, a document requirement, or a decline.

4.1 How the engine interrogates a fact

Question	How it is configured	Maria example
Present?	`facts:` list on each rule; absent fact → `on_missing`	No roof year at all → REFER
Evidence strong enough?	`min_evidence` on a ladder: `AGENT_REPORTED` < `CUSTOMER_DOC` < `VENDOR_MODEL` < `PUBLIC_RECORD` < `INSPECTION`	Agent says roof is 2019; rule needs `VENDOR_MODEL` or better, so ZestyAI or a BuildFax permit must back it

Question	How it is configured	Maria example
Fresh?	`max_age_days` per fact	OFAC result older than the configured window → re-screen before bind
Sources agree?	Corroboration rules compare the same fact across sources within a tolerance parameter	Agent 2019 vs permit 2008 → REFER with both values shown
Vendor down?	Adapter returns `UNAVAILABLE`, treated as missing	Blocking rules (OFAC, flood mandatory purchase) can never resolve to PASS

4.2 Anatomy of a rule

Predicates are written in **CEL** (Common Expression Language, `cel-go`): sandboxed, side-effect-free, no loops, deterministic, and native to Go. Numbers live in a separate **parameters** table so most changes never touch an expression. Matrix thresholds (e.g. max roof age by roof material × territory) use **decision tables**.

```
id: RSK-001
name: Shingle roof too old
checkpoint: RISK_ELIGIBILITY
scope: {carrier_paper: SPIC, product: HO3, state: FL, market_type: "*",
peril_scope: ALL_PERILS}
effective_from: 2026-11-01
facts:
  roof.material: {min_evidence: AGENT_REPORTED}
  roof.year:     {min_evidence: VENDOR_MODEL, max_age_days: 365, on_missing:
REFER}
when: >
  roof.material == "ASPHALT_SHINGLE" &&
  quote.effective_year - roof.year > param("roof.max_age.shingle")
outcome: KNOCKOUT
blocking: false
reason_code: ROOF_AGE_EXCEEDED
message: "Shingle roof is {{quote.effective_year - roof.year}} years old;
limit is {{param('roof.max_age.shingle')}}."
```

4.3 Outcomes and what the workflow does with them

Severity order: **KNOCKOUT > REFER > REQUIRE > FORCE_TERMS > INFO > PASS**. Every non-PASS result is kept (not only the worst), because each REFER or REQUIRE becomes its own stipulation.

Outcome	Effect on the quote	Cleared by
PASS	Nothing	—
INFO	Note on the quote, visible internally	—
FORCE_TERMS	Workflow applies a term (e.g. hurricane deductible floor) and re-rates; agent sees why	Automatic
REQUIRE (`stage: PRE_BIND` or `POST_BIND`)	Stipulation for a document/inspection; `PRE_BIND` locks Bind	Evidence arrives → re-evaluation passes, or a human clears it
REFER	Stipulation + `UnderwritingReviewWorkflow` child (*reuse*); Bind locked	Human decision via signal `uw_decision_recorded`
KNOCKOUT	Quote → `DECLINED` with reason code; agent sees the plain-English message	Terminal (a new quote after data correction)

4.4 Versioning, publishing and "no code deploy"

- **Immutable versions.** A `rule_set_version` is frozen once published; changes create a new version with its own `effective_from`.
- **Pinning.** A quote pins the version in force at creation for its validity period (same as the rate version), so an agent's quote doesn't shift under them. Rules flagged `applies_immediately` (e.g. a binding moratorium) bypass pinning.
- **Lifecycle:** `DRAFT → SIMULATED → APPROVED (four-eyes) → PUBLISHED → RETIRED`.
- **Simulate before publish.** Because every evaluation stored its fact snapshot, a candidate version can be replayed against the last N quotes to show the impact ("+42 knockouts, −18 referrals") before anyone approves it.
- **Reproducible.** Each `rule_evaluation` row stores version, fact-snapshot hash, results and actor, so "why was this quote referred in November?" is answerable years later (non-negotiable #5).
- **Binding moratorium as config.** Florida carriers routinely suspend new binding in counties under a hurricane watch or warning (industry practice, not in the Blueprint). Here that's one parameter, `moratorium.counties`, that Ops sets in minutes.

What does *not* belong in the rules engine: rating factors (rate tables in `rating`), taxes and fees (`product.tax_fee_rule`), coverage min/max limits (`CoverageDefinition`), notice periods (`notice_rule_set`). Rules decide eligibility and terms; they never compute money.

4.5 Maria's rule set (`SPIC × HO3 × FL`)

Thresholds shown come from About-SafePoint where it states them; everything else is a named parameter whose value Underwriting must supply.

ID	CP	Rule in plain words	Facts (min evidence)	Outcome
OFR-001	1	Offering open for the five-axis key on the effective date	product offering	KNOCKOUT
LOC-001	2	Geocode must be rooftop or parcel precision	geocode precision (VENDOR)	REFER
LOC-002	2	Distance to coast < `coast.knockout_miles`	coast distance (VENDOR)	KNOCKOUT
LOC-003	2	Distance to coast < `coast.force_ded_miles`	coast distance (VENDOR)	FORCE_TERMS: hurricane deductible ⩾ `hurr_ded.min_pct`
LOC-004	2	Protection Class 9 or 10	PPC (VENDOR, Verisk)	KNOCKOUT
LOC-005	2	Special Flood Hazard Area and a mortgagee on the quote	flood zone (certified determination), mortgagee party	REQUIRE `PRE_BIND`, blocking: flood evidence or lender notice
RSK-001	3	Shingle roof older than `roof.max_age.shingle` = 15	roof year (VENDOR_MODEL)	KNOCKOUT
RSK-002	3	Roof year sources differ by > `roof.tolerance_years`	roof year × 2 sources	REFER
RSK-003	3	Roof condition score below `zesty.min_roof_score`	ZestyAI roof score	REFER
RSK-004	3	Any prior loss > $0 in `clue.lookback_years`	CLUE (VENDOR)	REFER
RSK-005	3	Lapse in prior coverage	prior-carrier data (VENDOR)	REFER
RSK-006	3	Wind-mitigation credit claimed	OIR-B1-1802 form	REQUIRE `POST_BIND` (credit removed if not verified)

ID	CP	Rule in plain words	Facts (min evidence)	Outcome
COV-001	4	Cov A below `cov_a.min_pct_of_rc` of replacement cost	Verisk RC estimate	FORCE_TERMS or REFER
GAT-001	5	OFAC screen must be CLEAR and fresh	OFAC result (VENDOR, `max_age_days`)	REFER to OFAC review, **blocking**
GAT-002	5	County under binding moratorium	`moratorium.counties`	REFER, `applies_immediately`
ISS-001	6	Payment cleared (or escrow allowed), documents generated and QA'd	billing + document status	REQUIRE, blocking
PBV-001	7	Inspection result received within `inspection.due_days` = 30	inspection report	REQUIRE → on expiry per D-11
PBV-002	7	Inspected features differ from the features rated	inspection vs quote facts	REFER → endorsement re-rate

5. Domain entities & schemas

Eight schemas carry the journey; the two tables that make the design work are `quote.risk_fact` (facts with provenance, which the rules engine interrogates) and `policy.policy_transaction` (the bi-temporal legal record). All money is `BIGINT` cents; all config is effective-dated and scoped by the five-axis key.

Schema	Table	Key columns	Notes
`policy`	`party`	id, party_type (PERSON / ORG / TRUST), legal_name, identity_ref	Match-or-create at step 1
`policy`	`mortgagee_org`	id, name, loss-payee clause, escrow_supported, notice_prefs	Lender registry (seeded first, admin screen later)
`policy`	`policy_party`	policy_id, party_id, role, lien_position, loan_number, is_bill_to	Roles: NAMED_INSURED, ADDITIONAL_INSURED, MORTGAGEE, PRODUCER

Schema	Table	Key columns	Notes
`policy`	`policy` (**extend**)	+ policy_number, carrier_paper, product, form, state, market_type, peril_scope, product_version_id, status, source_quote_id	Additive columns; renewal slice keeps working
`policy`	`policy_term`	policy_id, term_no, effective_from, expiration, rate_version_id, ruleset_version_id	One row per term
`policy`	`policy_transaction`	txn_type, seq, effective_from/to, transacted_at, actor, premium_cents, snapshot (JSONB), reverses_txn_id	Bi-temporal; corrections = reversal + reissue
`policy`	`policy_number_seq`, `outbox_event`	per carrier/state counter; event_type, payload, published_at	Outbox written in the same transaction as the policy row
`product`	`carrier_paper`, `offering`	five-axis key, effective_from/to, status	CP1 reads this
`product`	`product_version`, `form`, `coverage_definition`	code, required, min/max/default limit cents, deductible_applicable	Coverage as a matrix, not form fields
`product`	`tax_fee_rule`	code, basis (FLAT / PCT), amount, payee (CARRIER / MGA / STATE), effective dates	Admitted vs E&S is just different rows

Schema	Table	Key columns	Notes
`rules`	`rule_set`, `rule_set_version`	checkpoint, scope, version, status, effective_from, content (JSONB), content_hash, approved_by	Immutable once published
`rules`	`rule_parameter`, `decision_table`	key, scope, value / rows	Where thresholds live
`rules`	`rule_evaluation`, `rule_result`	subject, checkpoint, version, fact_snapshot + hash; rule_id, outcome, reason_code, blocking, stage, terms	The audit and simulation source
`quote`	`quote`	quote_number, offering_id, status, effective_date, producer_id, pinned rule-set and rate versions, workflow_id, valid_until	The "Quote DB"
`quote`	`quote_location`, `quote_building`, `wind_mitigation`	address, lat/long, territory; construction, roof; the six OIR-B1-1802 features	Agent-entered values
`quote`	`risk_fact`	fact_key, value, evidence_level, source, source_ref, confidence, fetched_at, superseded_by	Every enriched or entered value
`quote`	`quote_coverage`	code, limit_cents, deductible, selected, forced_by_rule	Instances of `coverage_definition`
`quote`	`stipulation`	rule_id, type (REFER / REQUIRE), stage, blocking, status, assignee, due_at, cleared_by, rationale	Also used post-bind against `policy_id`

Schema	Table	Key columns	Notes
`quote`	`rating_result`, `premium_line`	rate_version, input_hash, technical/net/total cents, dirty; line_type, peril, amount_cents, visibility (INTERNAL / AGENT)	Visibility enforces the two-views rule
`rating`	`rate_version`, `rate_table`	scope, filing_ref, effective_from; table_key, dimensions, rows	Loaded from the filed rate manual
`billing`	`billing_account`, `invoice`, `invoice_line`	payer_type, payer_party_id, plan; line_type, amount_cents, revenue_owner	Records the policy-fee vs premium split
`billing`	`payment`, `gateway_event`	amount_cents, method, gateway_ref, status, idempotency_key; dedupe_key, raw payload	Webhooks deduped here
`document`	`document_template`, `policy_document`, `delivery_record`	scope, form_code; txn_id, template_version, hash, qa_status; recipient_role, channel, status	Delivery tracked per recipient
`integ`	`vendor_call`, `webhook_inbox`	adapter, request_hash, status, latency_ms, degradation_class, correlation_id	Raw vendor evidence for audit

The two load-bearing tables, as a starting point for the migrations:

```
-- quote/database/migrations/0001_init.sql (excerpt)
CREATE TABLE quote.risk_fact (
  id              UUID PRIMARY KEY,
  quote_id        UUID NOT NULL REFERENCES quote.quote(id),
  fact_key        TEXT NOT NULL,          -- e.g. 'roof.year',
```

```
'location.ppc', 'location.coast_distance_mi'
  value           JSONB NOT NULL,
  evidence_level  TEXT NOT NULL CHECK (evidence_level IN

('AGENT_REPORTED','CUSTOMER_DOC','VENDOR_MODEL','PUBLIC_RECORD','INSPECTION')
),
  source          TEXT NOT NULL,          -- 'agent', 'zestyai', 'buildfax',
'verisk', ...
  source_ref      TEXT,                   -- integ.vendor_call id or document
id
  confidence      NUMERIC(4,3),
  fetched_at      TIMESTAMPTZ NOT NULL,
  superseded_by   UUID REFERENCES quote.risk_fact(id)
);
CREATE INDEX ON quote.risk_fact (quote_id, fact_key) WHERE superseded_by IS
NULL;

-- policy/database/migrations/0002_policy_core.sql (excerpt, additive)
CREATE TABLE policy.policy_transaction (
  id              UUID PRIMARY KEY,
  policy_id       UUID NOT NULL REFERENCES policy.policy(id),
  term_id         UUID NOT NULL REFERENCES policy.policy_term(id),
  txn_type        TEXT NOT NULL CHECK (txn_type IN
('NB','END','REN','CAN','ASM','REV')),
  seq             INT  NOT NULL,
  effective_from  DATE NOT NULL,
  effective_to    DATE,
  transacted_at   TIMESTAMPTZ NOT NULL,
  actor           TEXT NOT NULL,          -- workflow id + activity, never an
API handler
  premium_cents   BIGINT NOT NULL,
  snapshot        JSONB NOT NULL,         -- the legal snapshot (Blueprint
6.7)
  reverses_txn_id UUID REFERENCES policy.policy_transaction(id),
  UNIQUE (policy_id, seq)
);
```

Migration rules: the `policy` module's next file is `0002_policy_core.sql` and is strictly additive (the renewal slice reads the existing columns); each new module starts at `0001_init.sql`; `schema.sql` is updated by hand alongside each.

6. Temporal workflows

Four new workflows plus one reused one carry Maria from draft to in-force; each is the only writer of the statuses it owns, and every vendor call, rule evaluation and clock read happens in an activity. All run on the existing `acme-property-us-policy-tasks` queue for now.

Workflow	ID / started by	Lifetime	Signals	Updates & queries
`QuoteWorkflow`	`quote-{quoteId}` ; `quote` module on create	Until accepted, declined, withdrawn or expired (quote-validity days, config)	`section_updated` , `quote_accepted` , `quote_withdrawn` , `stipulation_changed`	Update `RerateQuote` (sync); queries `getQuoteState` , `getReadiness`
`UnderwritingReviewWorkflow` (*reuse*)	Child of `QuoteWorkflow` , one per REFER	Until a human decides	`uw_decision_recorded`	—
`BindWorkflow`	`bind-{quoteId}` ; `QuoteWorkflow` on accept	Minutes (card) to days (payment pending / OFAC review)	`payment_result` , `ofac_review_decided`	`getBindState`
`PolicyLifecycleWorkflow`	`policy-{policyId}` ; `BindWorkflow` , **detached** (`ParentClosePolicy: ABANDON`)	The life of the policy	`documents_delivered` (now); endorsement, payment_failed, cancellation later	`getPolicyState`
`InspectionFollowUpWorkflow`	Child of `PolicyLifecycleWorkflow`	Until the inspection is verified or the timer expires	`inspection_result_received`	—

6.1 QuoteWorkflow

1. **Open:** `ResolveOffering` → `EvaluateCheckpoint(CP1)` ; KNOCKOUT → `SetQuoteStatus(DECLINED)` and end.
2. **On `section_updated(location)` :** `Geocode` , then in parallel `LookupProtectionClass` , `DetermineFloodZone` , `ComputeCoastDistance` , `EstimateReplacementCost` → `SaveFacts` → `EvaluateCheckpoint(CP2)` → `RangeRate` (returns a `PROJECTED` band).
3. **On `section_updated(building)` :** in parallel `GetRoofIntel` (ZestyAI), `GetPermits` (BuildFax), `PullCLUE` (timing per D-09) → `SaveFacts` → `EvaluateCheckpoint(CP3)` .

4. **Update `RerateQuote`:** `ApplyForcedTerms` → `EvaluateCheckpoint(CP4)` → `Rate` → `SaveRatingResult` → `SetQuoteStatus(QUOTED)`; returns the breakdown to the caller in one round trip. Any later material change marks the result `dirty` and requires another `RerateQuote`.
5. **REFER / REQUIRE results:** `OpenStipulations`; each REFER starts a child `UnderwritingReviewWorkflow` (generalised from `renewal_case` to `{subject_type, subject_id}`); Bind stays locked while any `PRE_BIND` blocking stipulation is open.
6. **On `quote_accepted`:** re-check readiness → `SetQuoteStatus(ACCEPTED)` → start `BindWorkflow` → await its result → `BOUND` ends this workflow; a failed bind restores `QUOTED`.
7. **Timer `valid_until`:** → `EXPIRED`.

6.2 BindWorkflow (saga; every step has a compensation)

1. **Pre-bind gates (CP5):** `ScreenOFAC` — a partial match opens the OFAC review and waits on `ofac_review_decided`; a timeout never proceeds. Re-run rules flagged `applies_immediately` (moratorium). Confirm no open blocking stipulations.
2. **Billing:** `CreateBillingAccount`, `CreateInvoice` (lines from `rating_result` + `tax_fee_rule`, split into premium vs policy fee). *Compensate:* `VoidInvoice`.
3. **Payment:** `CollectPayment` via `integ` → wait for `payment_result` (webhook relayed from `integ`). Failed or timed out → retry, then refer; **never bind unpaid**. *Compensate:* `RefundPayment`.
4. **Bind (single writer):** one activity, one DB transaction: `AllocatePolicyNumber`, create `policy`, `policy_term`, `policy_party`, NB `policy_transaction`, status `BOUND`, outbox `policy.bound`. *Compensate:* a `REV` transaction, never a delete.
5. **Start `PolicyLifecycleWorkflow`** detached.
6. **Issuance (CP6):** `GenerateDocuments` → `DocumentQA` (advisory, fire-and-forget) → `Deliver` to insured, agent and mortgagee → status `ISSUED`, outbox `policy.issued`. If issuance fails, the policy stays `BOUND` with an issuance stipulation — payment is not rolled back (payment ≠ issuance).

Escrow variant: when the bill-to party is the mortgagee, step 3 becomes `SendMortgageeInvoice`, and bind proceeds on an unpaid invoice only if the offering's `bind_on_escrow_invoice` flag is true (D-06).

6.3 PolicyLifecycleWorkflow and InspectionFollowUpWorkflow

- Lifecycle moves `ISSUED → IN_FORCE` on `documents_delivered`, and starts `InspectionFollowUpWorkflow` when any `POST_BIND` stipulation exists (Maria's RSK-006). Endorsement, arrears, cancellation and `CONTINUE_AS_NEW` at renewal are the

next plan, not this one.

- InspectionFollowUp: timer `inspection.order_delay_days` → `OrderInspection` → timer `inspection.due_days` (30) racing `inspection_result_received`.
  - Result arrives → `SaveFacts` (evidence `INSPECTION`) → `EvaluateCheckpoint(CP7)`: match → stipulation satisfied; mismatch → REFER, which will drive a re-rate endorsement.
  - Timer expires → the configured action, not a hard-coded one: `REMOVE_CREDIT_AND_REPRICE` or `START_UW_CANCELLATION` with state notice rules. About-SafePoint says "suspend", but Florida has no suspended-coverage status and the Blueprint state machine has none either (D-11).

Determinism checklist: no `http`, `time.Now()` or `rand` in workflow code; `workflow.Now` only for timers; every workflow and activity registered in `wfc/workflows/register.go`; search attributes `MycelState` / `MycelPhase` set at each phase so `wfm` can filter. If vendor latency starts starving renewal pollers, split an `...-enrichment` task queue — not before.

7. Integrations (`integ` adapters)

Eleven adapters are needed for Maria's path, and each one declares up front what happens when its vendor is down; only OFAC and flood determination (on a mortgaged home) are allowed to stop the line. Vendors named in About-SafePoint are used; where the docs name none, the vendor is an open choice.

Adapter	Vendor	Step	Facts produced (evidence level)	If the vendor fails
Address & geocode	Open	2	lat/long, precision, county (VENDOR_MODEL)	**Degrade** — zip centroid; LOC-001 refers
Protection class, BCEGS, replacement cost	Verisk / ISO	2	PPC, BCEGS, RC estimate (VENDOR_MODEL)	**Degrade** — missing fact → rule's `on_missing`
Flood zone determination	Open	2	zone, SFHA flag, determination id (PUBLIC_RECORD)	**Blocking** when a mortgagee exists; else Degrade

Adapter	Vendor	Step	Facts produced (evidence level)	If the vendor fails
Coast distance, wind pool, territory	In-house GIS on configured coastline + territory polygons, or a vendor (open)	2	distance (mi), wind-pool flag, territory code	**Degrade** — REFER
Roof intelligence	ZestyAI (Digital Roof, Z-PROPERTY)	3	estimated roof age, condition score (VENDOR_MODEL)	**Refer**
Building permits	BuildFax	3	re-roof permit year (PUBLIC_RECORD)	**Refer**
Loss history, identity, prior carrier	LexisNexis (CLUE)	1 / 3	prior claims, lapse (VENDOR_MODEL)	**Refer**
Sanctions / OFAC	Open	6	match status + score	**Blocking** — never proceeds on timeout
Payment gateway	Open	6	capture result, gateway ref (webhook)	**Retry-then-refer** — never binds unpaid
Document generation & delivery	Open	7	PDF + hash, delivery status per recipient	**Retry-then-refer**
Wind-mitigation inspection	Open	Post-bind	the six OIR-B1-1802 features (INSPECTION)	**Retry-then-refer**

Deferred: Cape Analytics aerial imagery (Degrade, useful later for CP3), ACORD/AL3 download (D-07), AI personas (fire-and-forget).

One contract for every adapter:

- Returns a normalised `FactSet {facts[], evidence_level, source_ref, status: OK | PARTIAL | UNAVAILABLE}` — the domain never sees a vendor payload.
- Hand-rolled context timeouts, bounded retries with backoff and a circuit breaker (Istio isn't deployed), a readiness check, and host-bound credentials.
- Every call written to `integ.vendor_call` with `correlation_id` = the workflow ID, so any fact traces back to the raw response.

- Property data cached per address for a configured number of days, so re-quotes don't re-buy the same report.
- **Inbound webhooks** (payment, inspection) land in `integ.webhook_inbox`, deduped by key, then go to the owning domain module (`billing`, `policy`), which records them and relays the signal to the workflow. As a backstop, `BindWorkflow` polls `QueryPaymentStatus` if no webhook arrives within its timeout.
- Payment calls pass our idempotency key to the gateway, so a retried activity cannot double-charge Maria.

8. UX screens — the minimum set

Of the ~50 screens in the UX Inventory, Maria's journey needs **7 of them plus 1 new Rules Manager**; six more fold in as tabs, and the rest are deferred with a stated stop-gap. All new screens are Svelte 5 (runes) inside `<policy-app>`, call the BFF only, and reuse Acme composite patterns.

8.1 Build

#	Screen	Why Maria's journey needs it	Acme pattern	Phase
1	**New Submission (Quote Wizard)** — 7 tabs	The capture flow itself (Blueprint 6.1–6.7)	Detail/Case tabs; Financial/Estimate on tab 5	1–5, tab by tab
2	**Quote & Bind Workbench**	Home for a saved quote; Bind Readiness rail; Request Info / Escalate / Bind	Detail/Case (main + 320px rail)	2 (shell), 4 (bind)
3	**Stipulation Queue**	Where REFER and REQUIRE outcomes land (Coastal Filter Tier 2)	Queue/Worklist	2
4	**Stipulation Clearance Workbench**	A human clears a referral; the Clear action signals the workflow	Review (AI findings → human decision)	2

#	Screen	Why Maria's journey needs it	Acme pattern	Phase
5	**Sanctions / OFAC Screening Review**	A partial match has to be clearable, or bind dead-ends	Review, no AI	4
6	**Submission Queue**	Agents resume drafts and see referred / declined quotes	Queue/Worklist	4
7	**Policy 360**	Confirms the issued policy: parties, coverages, documents, billing	Detail/Case	5
8	**Rules Manager** (*new — not in the Inventory*)	Makes "change rules without a deploy" self-service: edit parameters, simulate, approve, publish	Queue of rule sets + Detail with simulation panel	6 (publish API exists from phase 1)

8.2 Fold in as a tab (no separate screen yet)

- **Quote & Rating Detail (internal)** → role-gated "Rating" tab in the Quote & Bind Workbench; the server only returns `INTERNAL` premium lines to internal roles.
- **Document Center**, **Transaction History & Audit Trail**, **Billing Account / Invoice Detail** → tabs of Policy 360.
- **Payment Plan Configurator** → wizard tab 6 (full pay now, installments later).
- **Customer/Party 360** and **Mortgagee Management** → a match-or-create dialog and a lender lookup on wizard tab 1; the lender registry is seeded.

8.3 Deferred, with the stop-gap

Deferred screen	Covered meanwhile by
Rate Table Manager, Coverage & Product Configurator, Carrier/Paper config, Discount/Credit config, Range Rating config	Versioned seed migrations and an import job — still configuration, not code

Deferred screen	Covered meanwhile by
Payment Gateway Reconciliation	Stuck payments surface as BindWorkflow referrals; ops use `wfm` + the gateway portal (build in phase 6 hardening)
Flood & Mandatory Purchase Monitor, Declined & Withdrawn	Filters on the Stipulation and Submission queues
Integration Health Dashboard	Grafana over `integ.vendor_call` and traces
Dashboard, My Work, all servicing, assumption and reporting screens	Out of scope for this journey

8.4 What each wizard tab must show

1. **Customer & Policy** — party match-or-create; mortgagee lookup with lien order; effective date; resolved carrier paper and market type as read-only chips.
2. **Location** — enrichment results, each with a `ProvenanceBadge` (vendor vs agent); the low–high band labelled `PROJECTED`; the CP2 outcome banner.
3. **Building / Risk** — construction and roof, with the agent's value shown beside vendor and permit values; the six wind-mitigation features plus OIR-B1-1802 upload; CLUE losses read-only.
4. **Coverages** — matrix rendered from `CoverageDefinition`; forced terms shown locked with the rule's reason.
5. **Pricing Summary** — agent view (total, coverage-level premium, discounts, itemised taxes and fees); a `dirty` marker and Re-rate button after material changes; the delta shown explicitly.
6. **Payment** — payer (insured or mortgagee); card entry through the gateway's hosted fields, so card data never touches the PAS.
7. **Issuance** — live readiness checklist from `getReadiness`; Bind disabled with the blocking reasons listed; after bind, policy number and per-recipient delivery status.

Rule messages have two views too. Each rule carries an `agent_message` (plain English, no thresholds) and an `internal_message` (thresholds, fact values, sources); the BFF returns only the one the caller's role may see.

9. BFF API surface

Fourteen client routes and three aggregation endpoints cover the journey; every write requires an `Idempotency-Key`, and the BFF computes nothing — it routes, aggregates and rate-limits. Vendor webhooks bypass the BFF and go straight to `integ`.

Route	Upstream	Kind	Notes
`POST /v1/submissions`	`quote`	Write	Creates quote, starts `QuoteWorkflow`
`GET /v1/submissions`	`quote`	Read	Submission Queue (filters: status, agent, product)
`PATCH /v1/quotes/{id}/sections/{section}`	`quote`	Write	Saves draft data, sends `section_updated`
`POST /v1/quotes/{id}/rerate`	`quote` → Temporal Update	Write	Interactive rate-limit tier; returns breakdown per caller's role
`GET /v1/quotes/{id}/workbench`	`quote` + `rules` + `rating`	**Aggregation**	Header, tab data, stipulations, pinned versions
`GET /v1/quotes/{id}/readiness`	`quote` + `billing` + `document`	**Aggregation**	Drives the Bind Readiness rail and tab 7; partial-failure contract as in the renewal summary
`POST /v1/quotes/{id}/accept`	`quote`	Write	Sends `quote_accepted`
`POST /v1/quotes/{id}/payments`	`billing`	Write	Returns a hosted-payment session; key forwarded to the gateway
`GET /v1/stipulations`	`quote`	Read	Stipulation Queue
`POST /v1/stipulations/{id}/decision`	`quote`	Write	Clear / request info / escalate; relays `uw_decision_recorded`
`POST /v1/ofac-reviews/{id}/decision`	`quote`	Write	Relays `ofac_review_decided`; two-person rule configurable
`GET /v1/policies/{id}/360`	`policy` + `billing` + `document`	**Aggregation**	Policy 360 with documents, transactions and billing tabs
`GET/POST /v1/admin/rule-sets/{id}/versions`	`rules`	Read / write	Draft a new version, edit parameters
`POST /v1/admin/rule-sets/{id}/versions/{v}/simulate` (also `/approve`, `/publish`)	`rules`	Write	Admin tier; approve requires a second user

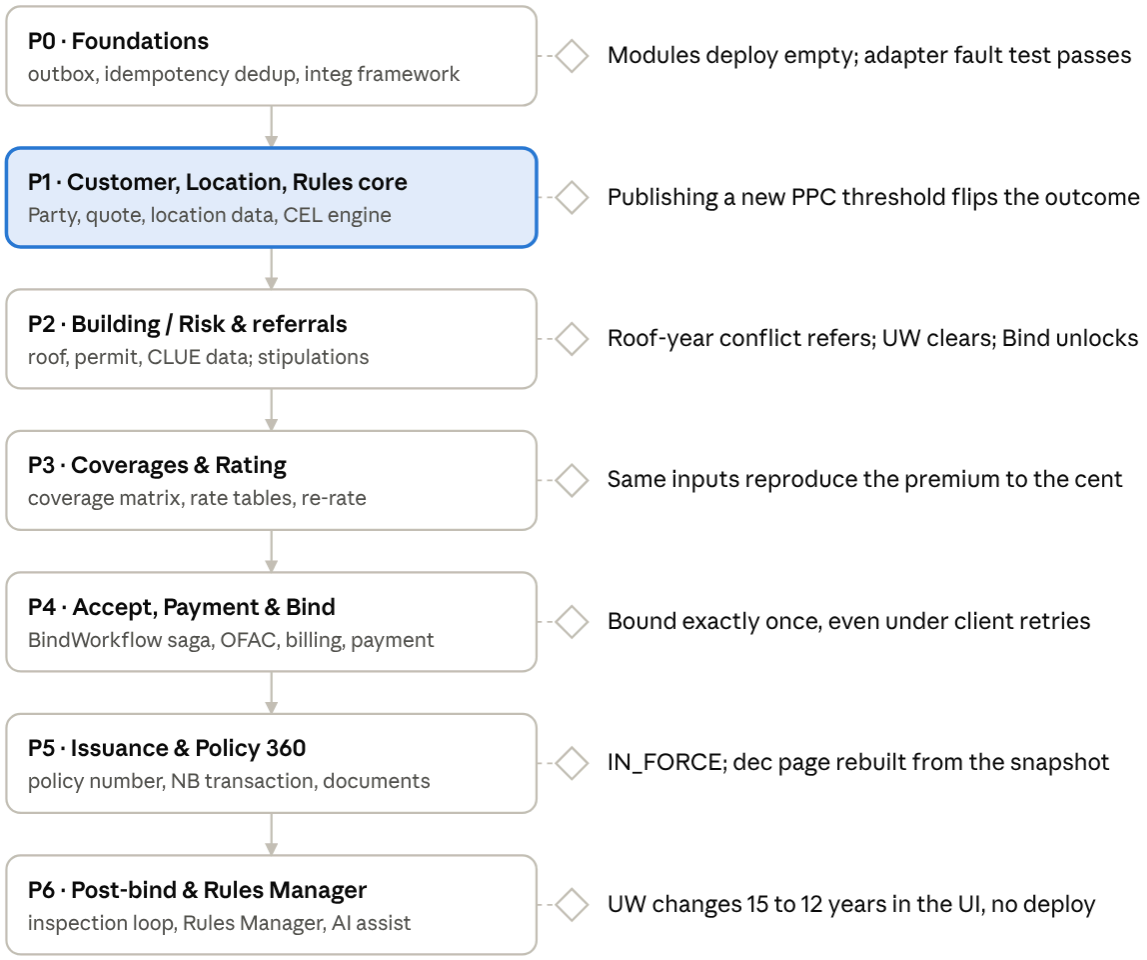
Prerequisite: idempotency **dedup storage** (today the BFF only enforces the header) must land before the payments route ships; otherwise a client retry double-submits a payment.

10. Phase-wise delivery plan

Seven phases follow Maria's journey in order, with the rules engine core landing in phase 1 so every later checkpoint is configuration from day one. Each phase ends with a demonstrable slice of her quote; durations are for the team to estimate.

The rules engine lands in phase 1, so every later phase is config-driven

Phases run in order; each ends at an exit gate (diamond) that must pass



Delivery roadmap · 7 phases, each with an exit gate

Each diamond is an exit gate: the phase is done only when that scenario passes end to end on the cluster (`mycel up --build`), not when its code merges.

Phase	Backend, workflows & rules	Integrations	Screens
P0 Foundations	Scaffold `product`, `rules`, `quote`, `rating`, `billing`, `document` modules; `policy` migration `0002` (party, term, transaction, outbox); money type (`int64` cents); seed `SPIC × HO3 × FL × ADMITTED × ALL_PERILS` offering; BFF idempotency dedup store	Adapter framework: `FactSet` contract, timeouts, retries, breaker, `vendor_call` log, webhook inbox, host-bound credentials	—
P1 Customer, Location, Rules core	Party match-or-create; quote shell; `QuoteWorkflow` steps 1–2; rules evaluator (CEL + parameters + decision tables + interrogation + evaluation log); publish API; CP1, CP2 rules; range rating (`PROJECTED`)	Geocode; Verisk PPC/BCEGS/RC; flood determination; coast distance / territory	Wizard tabs 1–2
P2 Building / Risk & referrals	Building + wind-mit entities; `risk_fact` corroboration; CP3 rules; stipulations; generalise `UnderwritingReviewWorkflow` to any subject	ZestyAI; BuildFax; LexisNexis CLUE	Wizard tab 3; Stipulation Queue; Stipulation Clearance Workbench; Workbench shell
P3 Coverages & Rating	`CoverageDefinition` matrix; rate-table import from the filed manual; pure `Rate()`; `RerateQuote` Update; tax/fee rules (admitted); CP4 rules; two-views authorization on premium lines	—	Wizard tabs 4–5; Workbench Rating tab (internal)
P4 Accept, Payment & Bind	`BindWorkflow` steps 1–5 with compensations; billing account, invoice, fee/premium split; CP5 gates; policy-number allocation; `PolicyLifecycleWorkflow` start	OFAC; payment gateway (hosted fields, webhook, status poll)	Wizard tab 6; OFAC Review; Submission Queue; Workbench Bind action
P5 Issuance & Policy 360	Issuance activities; legal snapshot on the NB transaction; templates by five-axis key; delivery records; CP6; outbox `policy.bound` / `policy.issued`	Document generation & delivery	Wizard tab 7; Policy 360 (documents, transactions, billing tabs)

Phase	Backend, workflows & rules	Integrations	Screens
P6 Post-bind & Rules Manager	`InspectionFollowUpWorkflow`; CP7 rules; expiry action per D-11; rule simulation (replay stored fact snapshots); four-eyes approve; AI personas (advisory only)	Wind-mit inspection vendor	Rules Manager; optional: `document_extraction_agent` on OIR-B1-1802 upload, `stipulation_triage_assistant`

Test focus per phase (hand to `safepoint-test-cases` for full plans): P1 — rule-version pinning and publish-without-deploy; P2 — vendor outage turns into REFER, never PASS; P3 — golden-file premium reproduction to the cent; P4 — duplicate webhook, activity retry after charge, OFAC timeout; P5 — issuance failure after payment leaves the policy `BOUND` with a stipulation; P6 — 30-day timer expiry and late-arriving inspection.

11. Non-negotiables and edge cases

All five Blueprint non-negotiables are enforced by structure in this plan, not by reviewer vigilance.

Non-negotiable	Where it bites here	How it is enforced
1. Deterministic, versioned rating, never an LLM	Tabs 4–5, `RerateQuote`	`rating.Rate()` is pure; rate version pinned; golden-file tests; AI personas only quote numbers the engine returned
2. Single writer	Every status change	Status-writing endpoints accept only the `wfc` service-account identity; widened to `quote.status`
3. Blocking gates never silently proceed	OFAC, flood mandatory purchase, surplus-lines diligent search (configured for E&S offerings, not triggered for Maria)	`blocking: true` rules cannot resolve to PASS on missing data or vendor timeout
4. Carrier/product/state is config	Rules, coverages, rates, taxes, templates	Five-axis key on every config table; code review rejects `if state == ...`

Non-negotiable	Where it bites here	How it is enforced
5. Explainable and reproducible	Every decision and number	`rule_evaluation` + fact snapshot; premium breakdown with rate version; `policy_transaction` actor and both dates; `vendor_call` by correlation ID

Edge cases this journey must handle:

- **Agent edits the roof year after `QUOTED`** → rating marked dirty, CP3 re-runs, the quote may move to REFER; the agent sees why.
- **Vendor contradicts agent** (permit 2008 vs entered 2019) → REFER showing both values and sources; never silently overwritten.
- **Rule set published mid-quote** → the quote keeps its pinned version, except `applies_immediately` rules such as a binding moratorium.
- **Storm moratorium between accept and payment** → `BindWorkflow` re-checks CP5 before charging.
- **OFAC before payment, deliberately** → a partial match stops the line before money moves, avoiding refunds.
- **Webhook lost or duplicated** → inbox dedupe plus status polling; one payment, one bind.
- **Issuance fails after payment** → policy stays `BOUND` with an issuance stipulation; no refund, no silent retry loop.
- **SFHA home with a mortgagee, flood written elsewhere (Equinox)** → REQUIRE proof of flood coverage or lender acknowledgment; the PAS doesn't write NFIP flood (D-01).
- **Two mortgagees** → lien order kept; both receive the dec page and are tracked per recipient.
- **Post-bind inspection disagrees with rated wind-mit features** → REFER, then a re-rate endorsement (additional or return premium) in the endorsement plan.
- **Inspection not back in 30 days** → configured action per D-11, never an undocumented "suspension".

12. Open decisions

Six existing register items gate parts of this plan, and seven new ones are proposed; D-10 and D-12 block phase 1, the rest can be settled before the phase that needs them.

ID	Decision	Blocks	Owner
D-01	Is NFIP flood in PAS scope? Drives LOC-005's REQUIRE wording	P1	Product
D-02	Default paper per state/product (Maria resolves to SPIC by config, never agent choice)	P1	Underwriting / Compliance
D-03	Notice-period table — needed only if D-11 picks cancellation	P6	Compliance
D-04	Wind-mit inspection validity period	P2 (RSK-006)	Underwriting / Pricing
D-06	Mortgagee escrow: bind on unpaid escrow invoice? (`bind_on_escrow_invoice`)	P4	Billing / Product
D-09	CLUE pull at quote or at bind (cost vs range-rating accuracy)	P2	Underwriting / Integration
D-10 (new)	Confirm eligibility/knockout rules are in PAS scope despite the Blueprint header; UW owns rule content, PAS owns the engine; who may approve a publish	**P1**	Underwriting + Product
D-11 (new)	Action when the post-bind inspection is late: remove credit and re-price, or start a UW cancellation (Florida has no "suspended" status)	P6	Underwriting / Compliance
D-12 (new)	Coastal thresholds: `coast.knockout_miles`, `coast.force_ded_miles`, `hurr_ded.min_pct` — About-SafePoint says only "< X miles"	**P1**	Underwriting
D-13 (new)	Vendors for geocode, flood determination, OFAC, payment gateway, documents and inspections	P0–P6	Integration / Procurement
D-14 (new)	Coexistence with Duck Creek OnDemand: does this path replace DC for FL HO3 new business, or run alongside it? Which system is the book of record?	Before go-live	Leadership / IT

ID	Decision	Blocks	Owner
D-15 (new)	Admitted FL line items for SPIC (premium tax treatment, EMPA surcharge, FIGA/Citizens assessments when levied) — the Blueprint itemizes only E&S	P3	Compliance
D-16 (new)	Rule-version pinning: at quote creation (proposed) or re-pinned at bind	P1	Underwriting